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Our team partnered with the UChicago Institute for Climate & Sustainable Growth through the Data Science Institute's research initiative AICE – AI for Climate. More specifically, we partnered with Rajat Masiwal, a postdoctoral scholar authoring a paper titled "Decision-oriented benchmarking to transform AI weather forecast access: Application to the Indian monsoon" (M26)[1]. In this paper, the authors aim to re-orient AI weather predictions by evaluating their accuracy with an eye towards farmer-focused metrics, like the timing of monsoon onset in local regions, which is highly pertinent to crop-planting but sometimes neglected in conventional meteorological benchmarking. We aimed to continue to develop the monsoon-bench codebase started in the Autumn 2025 quarter that can be published in tandem with the paper and used generally by interested stakeholders.

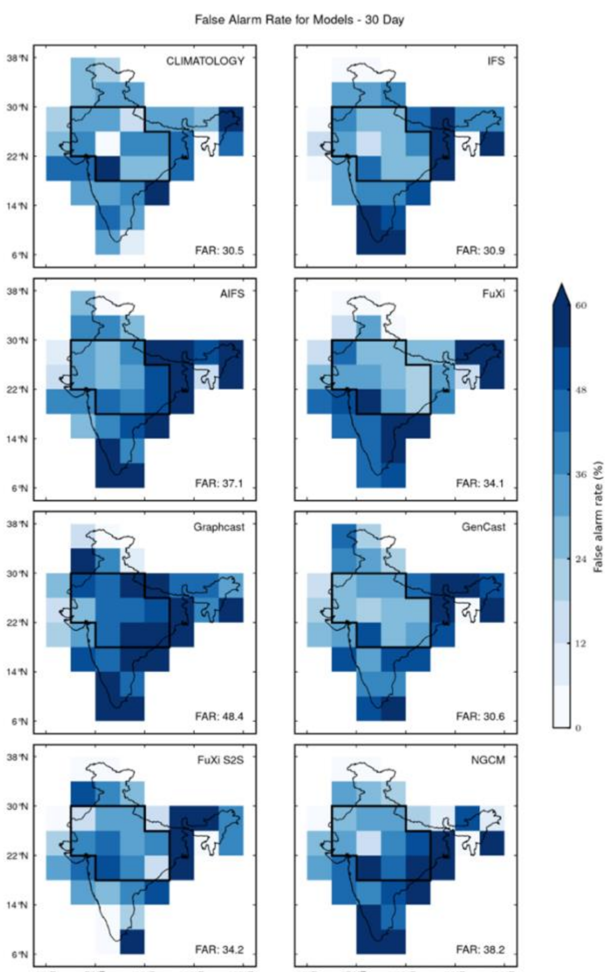

Figure 1: Map displaying the False Alarm Rate, defined as the percentage of monsoon onset predictions that are false positives, of 4x4 degree regions. Climatology predictions are based on weather data from previous years. IFS is not an AI model; it is a physics-based model. The models in the bottom three rows are all AI models.

Our primary task was to write code to reproduce the exact results and visualizations presented in the M26 paper. By recreating the paper's data pipeline – from data loading to metric calculation to spatial aggregation and visualization – we made the scientific results more reproducible. We used monsoon-bench as a unified interface to load, analyze, and visually summarize forecast performance of different models

An example output is displayed in Figure 1. Using the routines defined in monsoon-bench, the historical rainfall data and the predictions from 7 models were loaded and the metrics were calculated. The results were then plotted using visualization code originally written by Rajat but modified into a standard format usable in the monsoon-bench codebase.

Although we were able to reproduce most of the M26 paper visualizations, there are still some minor discrepancies in the metrics presented in the paper and the metrics calculated using the monsoon-bench code. Future work – such as parameter finetuning in the monsoon-bench metric calculation process – is still required to bridge the remaining gaps between Rajat's paper figures and the figures produced with the monsoon-bench code.

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[1] https://arxiv.org/abs/2602.03767